Improving Dating Outcomes by Predicting for Conversations at Hinge

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Abstract

The ultimate measure of success for a dating platform is helping people go on in-person dates. In practice, ranking systems for such platforms rely on observable proxies, such as likes and matches, to predict whether an in-person date will occur. This paper introduces *conversations*, when both users have sent at least one message, as a stronger signal of dating intent. We model the dating journey as a probabilistic chain, decomposing the likelihood of a conversation into three sequential stages: the probability of a like, a match given a like, and a conversation given a match. By targeting conversations as a late-stage observable proxy for dating intent, this approach captures a higher-fidelity signal than using likes and matches alone. We present the production deployment of this system at Hinge via a multi-task two-tower model that jointly predicts three objectives with a shared embedding layer. User interaction sequences are encoded with a transformer architecture to capture temporal patterns. Our A/B test results demonstrate a +2.8% improvement in number of conversations and a +1.4% increase in exchange coverage.

CCS Concepts

• **Information systems → Retrieval models and ranking**.

Keywords

Recommender Systems, Multi-task Learning, Reciprocal Recommendation

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1 Introduction

Dating platforms present a unique challenge in recommendation systems: positive outcomes require mutual interest between two users, not just one-sided engagement. Unlike traditional feed ranking systems that optimize for individual user satisfaction, dating recommendations must solve a reciprocal matching problem where outcomes depend on mutual satisfaction of both parties.

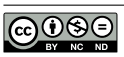
Hinge is a dating application with millions of active users. Users interact with a feed of potential profiles and send "likes" to indicate interest. When a recipient accepts an incoming like, the pair becomes a "match" and can start a conversation. If this conversation progresses positively, users typically share contact information and transition to scheduling an in-person date. While in-person dates represent true success, measuring them directly is challenging: they occur off-platform, rely on user-provided signals such as surveys to measure, and are sparser than in-app engagement. Many published systems therefore optimize for proximal signals such as likes and matches, which are easier to measure but may not align with users' goals of finding meaningful relationships.

This paper argues that conversations serve as a more effective proxy for in-person dates than top-of-funnel signals. The main contributions are:

- Formulating the dating funnel as a probabilistic chain that decomposes conversation likelihood into three conditional stages. We demonstrate that conversations provide a high-fidelity, observable proxy for in-person dating intent.
- A production multi-task two-tower ranking system deployed at Hinge, with online A/B test results demonstrating improvements across multiple measures of dating outcomes.

1.1 Terminology

An *impression* occurs when a profile is surfaced to a user in their feed, at which point the user either sends a like or skips to the next profile. The user who sends the like is the *initiator*, and the recipient is the *responder*, who can accept or skip. Accepting forms a *match* that enables messaging. When both users have sent at least one message, this is a *conversation*. When users coordinate to meet in person, whether by sharing contact information or scheduling directly in-app, this is *date planning*. The dating funnel is modeled as: impression → like → match → conversation → date planning → in-person date.



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2 Related Work

Dating platforms are an example of reciprocal recommendation systems, where successful outcomes require mutual interest between both parties [5]. Prior work on dating recommendations for production platforms has primarily focused on predicting observable proxies such as likes, matches, and time spent [3, 6, 8, 9]. While matches represent mutual interest, they remain an early-stage signal that does not fully capture whether users will progress toward meaningful connections. Our work extends this line of research by predicting for conversations, a later-stage signal that better approximates in-person dating intent.

Multi-task learning has become a standard approach for jointly modeling multiple objectives in recommendation systems. The Entire Space Multi-Task Model (ESMM) [4] addresses sample selection bias by modeling the entire user journey rather than conditioning only on observed events at each stage. Our work applies similar principles to the dating domain, modeling the full funnel from impressions through conversations.

Two-tower architectures have emerged as an efficient paradigm for large-scale retrieval and ranking [11]. These models use separate towers to encode users and items into a shared embedding space, enabling efficient nearest neighbor search at inference time. Deep & Cross Networks (DCN-V2) [10] extend this approach with explicit feature crossing layers that capture high-order feature interactions.

Transformer-based architectures have emerged as a powerful approach for modeling sequential user behavior in recommendation systems [1, 2]. Self-attention mechanisms enable these models to capture long-range dependencies in user interaction histories. Our model incorporates transformer encoders to process user behavior sequences, capturing temporal patterns in liking and matching behavior that inform predictions about future outcomes.

3 Approach

This section describes the multi-task learning approach for ranking candidates, justifies conversations as the prediction target, and details training and deployment.

3.1 Overall Architecture

Our recommendation system consists of two main stages: candidate retrieval and ranking. In the retrieval stage, we filter a large pool of profiles based on user preferences (age, distance, gender, etc.) and exclude previously seen profiles, producing a set of thousands of candidates. The ranking stage then orders these candidates using a multi-task two-tower model.

The proposed ranking model exploits the sequential structure of the dating funnel (like → match → conversation). Inspired by ESMM, the approach models the entire user journey rather than conditioning only on observed events at each stage. Like, match, and conversation are denoted as binary variables l, m, and c respectively, for a given impression $\boldsymbol{x}$. The goal is to predict the joint probability $p(l, m, c \mid \boldsymbol{x})$, shown in Eq. 1. This formulation addresses the data sparsity of conversations by training on the entire space of impressions, leveraging the more abundant like and match signals through the multiplicative structure:

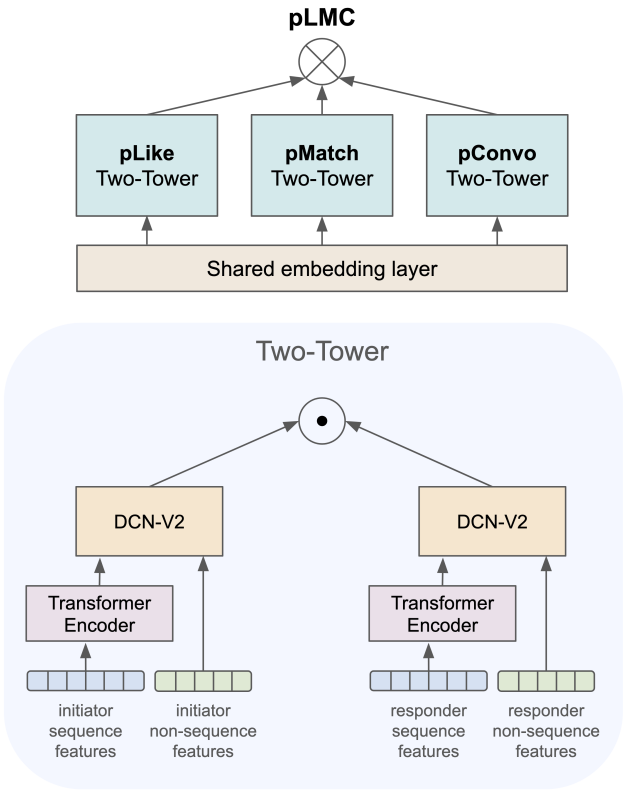


Figure 1: Architecture overview showing three two-tower modules with shared embeddings. Each module predicts one funnel stage: like (pLike), match given like (pMatch), and conversation given match (pConvo). The final ranking score pLMC represents the joint probability of all three events occurring.

$$\underbrace{p(l, m, c \mid \boldsymbol{x})}_{\text{pLMC}} = \underbrace{p(l \mid \boldsymbol{x})}_{\text{pLike}} \times \underbrace{p(m \mid l, \boldsymbol{x})}_{\text{pMatch}} \times \underbrace{p(c \mid m, \boldsymbol{x})}_{\text{pConvo}} \tag{1}$$

where pLike, pMatch, and pConvo represent the probabilities of like, match given like, and conversation given match, respectively.

Model Architecture. Our model consists of three two-tower modules (six towers total), one module per prediction head (pLike, pMatch, pConvo). Each tower uses a 5-layer Deep & Cross Network (DCN-V2) architecture with 32-dimensional embeddings. The three towers on each side share a common embedding layer to facilitate information transfer across tasks. To capture user preferences, we encode the sequence of profiles each user has previously interacted with using a self-attention transformer encoder with learnable positional embeddings. This implementation is shown in Figure 1.

3.2 Why Conversations?

Conversations serve as a more effective proxy for in-person dating intent than likes and matches. Because conversations occur deeper in the funnel, they signal more meaningful intent to go on a date while remaining directly observable. To validate this, we analyzed

historical data on the platform using periodic in-app surveys to measure in-person dates. Table 1 shows the relative likelihood of an in-person date occurring, conditioned on reaching different stages of the funnel.

Table 1: Relative likelihood of in-person dates at each funnel stage across U.S. users

Funnel Stage	Relative Lift
Like sent	1x (baseline)
Match formed	~14x
Conversation	~20x

Crucially, conditioning on a conversation provides substantially more signal about dating intent than conditioning on a match alone, making conversations a more effective optimization target than top-of-funnel metrics.

While one could argue for optimizing even further down the funnel, towards date planning or in-person dates, these later stages present significant challenges. Date planning depends heavily on how conversations progress, which cannot be observed at impression time since conversations only occur after users match. This makes these signals substantially sparser and harder to predict from pre-match features alone. In-person dates are difficult to observe reliably and typically occur days to weeks after the initial match, introducing temporal delay in the feedback signal. Conversations strike an optimal balance between signal quality and observability.

3.3 Training and Deployment

A separate ranking model is trained for each geographical region where Hinge operates. The model is trained with the Adam optimizer using a learning rate of 0.001 and a batch size of 1M examples. Training is done using distributed data parallel (DDP) across multiple GPUs with 16-bit mixed precision to reduce GPU memory.

During training, we use binary cross-entropy loss for each of the three prediction tasks. The final loss function is a weighted sum of the individual task losses:

$$\mathcal{L} = \mathcal{L}_{\text{like}} + \alpha\mathcal{L}_{\text{match}} + \beta\mathcal{L}_{\text{convo}} \tag{2}$$

where α and β are scalar weights tuned via hyperparameter search. $\mathcal{L}_{\text{like}}$ is computed on pLike, $\mathcal{L}_{\text{match}}$ on the product pLike $\times$ pMatch, and $\mathcal{L}_{\text{convo}}$ on pLMC. Gradients propagate through the multiplicative structure: $\mathcal{L}_{\text{match}}$ backpropagates through both pLike and pMatch, while $\mathcal{L}_{\text{convo}}$ backpropagates through all three prediction heads. This allows conversation and match labels to shape the learned representations of earlier stages, so that the model learns features relevant to downstream outcomes across the full funnel.

Each tower produces a single embedding vector, for six embeddings in total. Responder embeddings are precomputed and indexed in ElasticSearch, while initiator embeddings are stored in PostgreSQL and used as query vectors. At inference time, the three task-level dot products between initiator and responder embeddings are multiplied to produce the final ranking score pLMC.

Table 2: A/B test metric improvements using our conversation-optimized model

Metrics	Improvements
Likes	−0.4%[1]
Matches	+2.4%
Conversations	+2.8%
Exchange coverage	+1.4%

4 Results

This section evaluates the performance of the conversation-optimized ranking model using online A/B testing and analyzes its impact across the dating funnel.

4.1 Experimental Setup

The baseline model is another multi-task two-tower ranking model, but optimized for match probability rather than conversation probability. Instead of three prediction heads, the baseline model has two for pLike and pMatch, whose product serves as the ranking score.

Recommendations at Hinge are loaded in batches of 20 profiles. Four fixed positions in each batch are allocated for this experiment. In the treatment condition, these four positions are filled by the conversation-optimized model, while in the control condition they are filled by the baseline match-optimized model. The remaining 16 positions are filled by other models and remain constant across both conditions. This limited exposure allows for impact measurement while maintaining user experience stability.

The experiment is conducted as a metro-level randomized controlled test, with fifteen metros assigned to each variant over a three-week period. This design is chosen because user-level randomization is less reliable in this setting: recommendations in a two-sided marketplace can induce interference between units, weakening the stable unit treatment value assumption (SUTVA) that underlies standard A/B test inference. Concretely, if the treatment model surfaces better candidates, treated initiators send more and better-targeted likes. Because any given user has a finite amount of attention to distribute across incoming likes, this redistributes the likes and matches received by control users, contaminating the counterfactual and biasing the estimated effect on downstream metrics such as matches and conversations in either direction. Because the overwhelming majority of Hinge matches occur between users in the same metropolitan area, randomizing at the metro level approximates a collection of independent two-sided markets, one per experimental unit. This cluster-level design restores exchangeability between conditions and yields a valid estimate of the total treatment effect.

All user personal data is managed in conformance with all applicable privacy laws.

4.2 A/B Testing Results

Several key metrics are reported: *Likes*, *Matches*, and *Conversations*, which represent the average number of likes received, matches

[1]Not statistically significant. All other metrics are statistically significant.

Table 3: Relative conversion rate improvements across funnel stages of the proposed model vs. control

Metrics	Improvements
Impression to like rate	−20.4%
Like to match rate	+42.2%
Match to conversation rate	+6.9%

received, and conversations per user, respectively. *Exchange coverage* measures the percentage of unique users who shared contact information via messaging. While date planning encompasses both contact sharing and in-app scheduling, exchange coverage serves as a measurable proxy for date planning activity and, by extension, in-person dating intent.

Online test results are shown in Table 2. Significant improvements are observed for matches, conversations, and exchange coverage. The system successfully shifts interactions toward higher-intent outcomes without negatively impacting overall platform usage.

Table 3 reports relative changes in conversion efficiency between consecutive stages of the dating funnel, comparing the conversation-optimized model to the baseline model which optimizes for matches. These metrics are computed over the four slots per batch dedicated to the experiment. As expected, an improvement is observed in the conversion from matches to conversations, reflecting direct prediction of reciprocal messaging.

Notably, the model also produces a large increase in like-to-match efficiency, accompanied by a decrease in impression-to-like conversion. This behavior arises from the multiplicative structure of the ranking objective: because the final score is proportional to the product of predicted like, match, and conversation probabilities, candidates must score highly across all stages of the funnel to be ranked early. As a result, exposure shifts away from broadly likable profiles toward profiles with higher predicted mutual intent, reducing likes slightly while improving downstream dating outcomes.

4.3 Feature Importance Analysis

To understand the inputs that drive the conversation-optimized model's improvements, we compute per-feature attributions using integrated gradients [7]. We compute attributions on a held-out sample of impressions for both the conversation-optimized model and the baseline model, aggregate the absolute attributions within each feature category, and normalize so that category importances sum to 100% within each model.

Features are grouped into four categories: *profile features* describing self-reported user attributes, *preference features* describing desired attributes in candidates, *tenure & recency features* capturing user lifecycle and recency signals such as join date, and *behavioral features* summarizing user activity on the platform. Results are reported in Table 4.

The conversation-optimized model exhibits a reallocation of attribution, reducing reliance on behavioral features by 11 percentage points and shifting most of that mass toward tenure & recency features, which nearly double in aggregate importance. Profile and preference category importances are roughly unchanged. This reallocation is consistent with the structure of the conversation objective. Producing a conversation requires both users to be actively engaged on the platform, so lifecycle and recency signals carry more predictive value. This shift is also consistent with the funnel-level results in Table 3, where the model trades marginal likes for higher like-to-match and match-to-conversation efficiency.

Table 4: Relative feature importance by category. Δ denotes the change from baseline to the conversation-optimized model, in percentage points.

Category	Baseline	Conversation	Δ
Profile features	18.3%	19.4%	+1.1
Preference features	13.0%	13.5%	+0.5
Tenure & recency features	9.9%	19.3%	+9.4
Behavioral features	58.8%	47.8%	−11.0

5 Conclusions

This paper presented a multi-task two-tower ranking system that predicts for conversations as a proxy for in-person dating intent. By formulating the dating funnel as a probabilistic chain, the approach addresses sample selection bias and data sparsity inherent in predicting late-stage outcomes.

Beyond dating platforms, this work addresses fundamental challenges in reciprocal recommendation systems where success requires mutual selection between parties. The same funnel decomposition applies naturally to other two-sided settings, such as talent platforms and marketplaces with host or provider approval. Future work could explore alternative formulations of the ranking score beyond the multiplicative chain.

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Speaker Bio

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